CPIT-455 Syllabus

Catalog Description

CPIT-455 Software Engineering (II)
Credit: 3 (Theory: 3, Lab: 0, Practical: 1)
Prerequisite: CPIT-251
Classification: Elective

The objective of this course is to explore software advanced engineering principles and techniques used in the specification, design, and testing of software systems Topics include software engineering confirmation and authentication, methods and ways of software testing building and managing software development teams, scientific and practical methods for calculating the costs of software development, quality management in software development, processes development in software production, and reconstruction management in software production.

Class Schedule

Lab/Tutorial 90 minutes 1 times/week

Meet 50 minutes 3 times/week or 80 minutes 2 times/week

Textbook

Ian Sommerville, , "Software Engineering", Addison-Wesley; 10 edition (2015-03-24)

ISBN-13 9780133943030 **ISBN-10** 0133943038

Grade Distribution

Week	Assessment	Grade %
7	Homework Assignments	10
8	Exam 1	30
15	Group Project	20
15	Graded Lab Work	10
16	Exam	30

Last Articulated

October 12, 2023

Relationship to Student Outcomes

1	2	3	4	5	6
	x				x

Course Learning Outcomes (CLO)

By completion of the course the students should be able to

1. **Design dependability in systems by using redundancy and diversity. (2)**
2. **Write functional and non functional dependability and security requirements (2)**
3. Apply metrics for reliability specification to specify measurable reliability requirements. (2)
4. **Design safety in safety critical systems (2)**
5. Design and test secure systems (2)
6. Design systems to support resilience (2)
7. Identify the elements involved in the re-use of code and describe an approach to software reuse based on the composition of standardized, reusable components (6)
8. Describe an approach to software reuse based on the composition of standardized and reusable components (6)
9. Explain distributed systems engineering and distributed systems architectures (6)
10. Explain the idea of a system of systes and discuss the challenges of building complex systems of software systems (6)
11. Design and implementation of test cases in the different testing strategies (2)

Coordinator(s)

Dr. Mounira Taileb, Associate Professor

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Topics Coverage Durations

Topics	Weeks
Dependable systems	2
Reliability engineering	2
Safety engineering	1
Security engineering	2
Resiliene Engineering	2
Software Reuse	1
Distributed software engineering (Chapter 17)	1
Component-based software engineering (Chapter 16)	1
Systems of systems (Chapter 20)	1
Code coverage	2